Multiobjective Maintenance Planning Optimization for Deteriorating Bridges Considering Condition, Safety, and Life-Cycle Cost

Min Liu, A.M.ASCE,[1] and Dan M. Frangopol, F.ASCE[2]

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Abstract: Many of the currently available bridge management system tools focus on minimizing life-cycle maintenance cost of deteriorating bridges while imposing constraints on structural performance. The computed single optimal maintenance planning solution, however, may not necessarily meet a bridge manager's specific requirements on lifetime bridge performance. In this paper the life-cycle maintenance planning of deteriorating bridges is formulated as a multiobjective optimization problem that treats the lifetime condition and safety levels as well as life-cycle maintenance cost as separate objective functions. A multiobjective genetic algorithm is used as the search engine to automatically locate a large pool of different maintenance scenarios that exhibits an optimized tradeoff among conflicting objectives. This tradeoff provides improved opportunity for bridge managers to actively select the final maintenance scenario that most desirably balances life-cycle maintenance cost, condition, and safety levels of deteriorating bridges.

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Introduction

The economical allocation of limited funding for maintaining both safety and serviceability of deteriorating highway bridge networks constructed in the United States as well as other parts of the world necessitates suitable bridge maintenance system (BMS) tools that can maximize cost effectiveness. Available practical BMS software packages include *BRIDGIT* (Hawk and Small 1998) and *Pontis* (Thompson et al. 1998) systems that determine the timing and types of maintenance interventions based on discrete condition states resulting from visual inspections.

Numerical procedures have recently appeared in the literature that search for optimal maintenance planning. Most of the existing numerical procedures aim at minimizing the expected cumulative maintenance cost over a specified time horizon while treating other important bridge performance criteria as relevant constraints in the optimization process. For example, Kong and Frangopol (2003) presented a methodology for reliability-based life-cycle cost optimization of deteriorating bridges using the feasible search direction combined with the branching and multidimensional interpolation methods. They used the present value of cumulative maintenance cost during a specified service life as the single objective function. Constraints on system reliability can be enforced to ensure satisfactory lifetime safety and serviceability levels for deteriorating bridges (Estes and Frangopol 2001).

Despite the advantage of these automated maintenance planning procedures that can provide a rational solution based on life-cycle cost minimization, one of the arguments against such approaches is that the computed single optimal maintenance solution may not necessarily meet a bridge manager's specific requirements on bridge performances. As a remedy, the numerical algorithms could be rerun using another set of constraints that is compatible with specified performance demands of the bridge manager. Facing a single maintenance solution, a bridge manager still has no opportunity to make judgmental decisions upon the choice of final maintenance scenario from a group of alternative ones that lead to distinct bridge performances other than merely emphasizing the life-cycle cost optimization issue.

A more powerful approach is to consider all relevant bridge performance aspects as separate objective functions, resulting in a multiobjective optimization formulation for the bridge maintenance planning problem. Instead of a unique optimal maintenance solution as is obtained from the traditional life-cycle cost optimization approaches, a set of alternative maintenance solutions is searched that comprises the best possible tradeoff among all competing objectives under consideration. Bridge managers' valuable experiences as well as personal preference can then be incorporated into the decision-making process to determine the compromise maintenance scenario that can most desirably balance different objectives. Previous research efforts on multiobjective maintenance planning optimization include Liu et al. (1997) who minimized both maintenance cost and a normalized weighted average deterioration measure for maintaining bridge decks; Miyamoto et al. (2000) who investigated optimal maintenance of existing bridges by minimizing maintenance cost and maximizing bridge load-carrying capacity as well as durability; and Furuta et al. (2004) who simultaneously considered minimization of life-cycle cost, maximization of service life, and maximization of tar-

[1]Research Associate, Dept. of Civil, Environmental, and Architectural Engineering, Univ. of Colorado, Boulder, CO 80309-0428. E-mail: liu@colorado.edu

[2]Professor, Dept. of Civil, Environmental, and Architectural Engineering, Univ. of Colorado, Boulder, CO 80309-0428. E-mail: dan.frangopol@colorado.edu

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get safety level for maintaining civil infrastructure systems.

In this paper, life-cycle maintenance planning optimization of deteriorating bridges is formulated as a multiobjective problem to be solved by an evolutionary algorithm. Bridge condition index, safety index, and life-cycle maintenance cost are considered as three separate objective functions. Application examples using real data of reinforced concrete highway crossheads are provided to demonstrate the usefulness of the proposed procedure.

Multiobjective Optimization Framework

The present study aims to develop an automated procedure that produces optimal bridge maintenance solutions while accounting for multiple different performance measures. Three objectives are considered in the present optimization problem: (1) improving the worst condition index value, (2) improving the worst safety index value over the intended service life, and (3) reducing the present value of the life-cycle maintenance cost. The evaluation methods are also described. A genetic algorithm (GA) is used as the numerical tool for solving the posed multiobjective optimization problem, which is conceptually described by the flowchart in Fig. 1.

Condition and Safety Indices

The condition index used in this study is a continuous generalization (Neves and Frangopol 2005) of the visual inspection-based and discrete-valued quantities that are adopted in current bridge management systems. For example, Denton (personal communication 2002) assigns the following discrete condition values: 0=no chloride contamination, 1=onset of corrosion, 2=onset of cracking, and 3=loose concrete/significant delamination. Therefore, the larger value the condition index takes, the worse a bridge's condition is. The safety index in accordance with *DB12/01* (Highways Agency Standard for Bridge Assessment 2001) is quantified as the ratio of available to required live load-carrying capacity. The smaller value the safety index takes, the lower a bridge's safety level becomes.

A multilinear computational model (Frangopol et al. 2001) is used to depict deterioration profiles of structure performance in terms of condition and safety indices with and without maintenance interventions, respectively. As shown in Fig. 2, the performance profile of a deteriorating structure without maintenance is characterized by three parameters: initial performance index P_0, time to damage initiation T_I until which no deterioration is present, and a constant deterioration rate α. Mathematically

$$P(t) = \begin{cases} P_0 & \text{if } t < t_i \\ P_0 - \alpha t & \text{if } t \geq t_i \end{cases} \quad (1)$$

The effects of a time-based maintenance action on the structure performance profile are described by the first application time T_{PI}, time interval of subsequent cyclic application T_P, immediate performance improvement γ, time delay of deterioration T_D, reduction of deterioration rate δ, and duration of maintenance effect T_{PD} (Frangopol et al. 2001). The linear superposition rule is assumed applicable for computing the composite performance index profile of a structure with maintenance interventions (Kong and Frangopol 2003). Mathematically

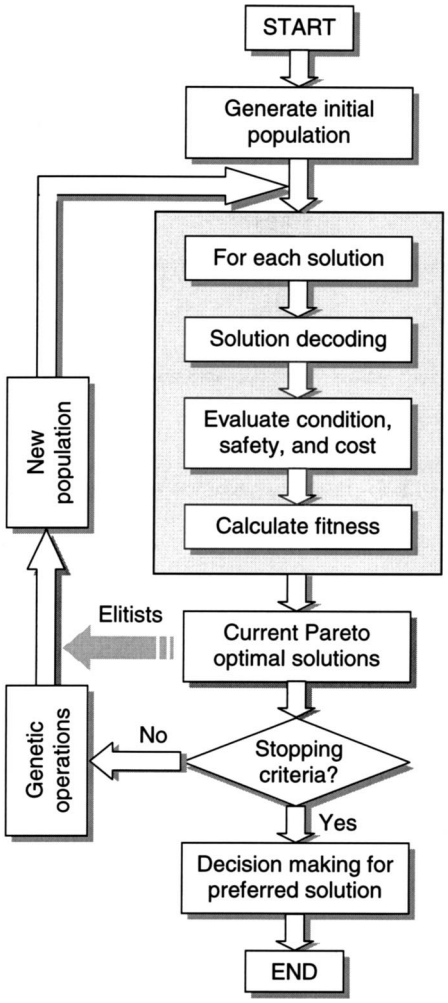


Fig. 1. Flowchart for genetic algorithm-based multiobjective bridge maintenance planning optimization

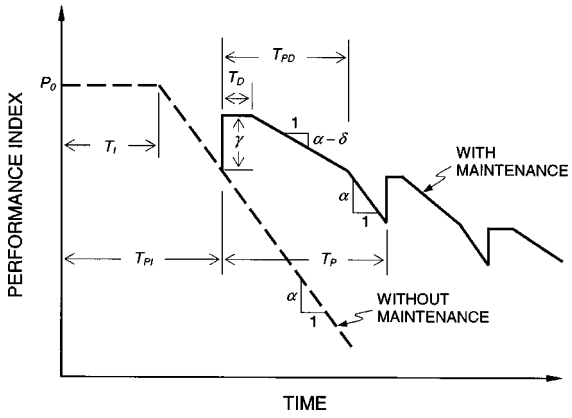


Fig. 2. Performance index profiles with and without maintenance interventions

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$$P(\tau) = \begin{cases} P_{(0)} + \gamma & \text{if } 0 \leq \tau < t_d \\ P_{(0)} + \gamma - \max(\alpha - \delta; 0) \cdot (\tau - t_d) & \text{if } t_d \leq \tau < t_{pd} \\ P_{(0)} + \gamma - \max(\alpha - \delta; 0) \cdot (t_{pd} - t_d) - \alpha(\tau - t_{pd}) & \text{if } \tau \geq t_{pd} \end{cases} \quad (2)$$

where τ = relative time variable since application of maintenance. The max $(\cdot)$ operator excludes the numerical possibility of a negative net deterioration rate. Generally speaking, if there are K maintenance interventions active at the time of interest, the instantaneous net deterioration rate ε is calculated as

$$\varepsilon = \max\left(\alpha - \sum_{i=1}^{K} \delta_i; 0\right) \quad (3)$$

In contrast, performance-based maintenance interventions are applied when a performance index (condition or safety) reaches a predefined target value. Therefore, their times of application cannot be determined a priori. Other defining parameters as specified above still apply.

Present Value of Cumulative Life-Cycle Maintenance Cost

Each maintenance intervention is associated with prescribed amount of monetary expenses that covers both direct and indirect costs. The effect of application time on the cost estimate can be accounted for by a discount rate that converts future cost into the value at a predefined reference time instant (usually the present time), i.e.

$$C_{\mathrm{PV}} = \sum_{i=1}^{N} \frac{C_{t_i}}{(1+\nu)^{t_i}} \quad (4)$$

where C_{PV} = present value of the cumulative (life-cycle) cost due to all maintenance interventions; N = number of maintenance interventions; C_{t_i} = cost of maintenance interventions applied at time t_i; and ν = discount rate.

Genetic Algorithms

Numerical algorithms are needed to automatically guide the search process for solution(s) that can optimize the predefined objective function(s). Different traditional algorithms are available for the existing bridge maintenance plannings that are formulated as optimization problems for life-cycle maintenance cost minimization only. These classical single-objective based algorithms, however, usually cannot directly handle multiple and conflicting objective functions such as the present problem where three distinct merit measures are treated separately as well as simultaneously in the optimization process.

The GAs are selected in this study as the primary optimization tool to solve the posed multiobjective optimal maintenance planning problem. The GAs are global stochastic search and optimization engines that mimic natural evolution and are based on Darwin's survival-of-the-fitness theory in the biological sciences (Goldberg 1989). They typically create improved solutions using three basic operators, i.e., selection, crossover, and mutation, until prescribed stopping criteria are considered met. Compared to most gradient-based traditional methods, GAs are problem-independent methods and do not rely on sensitivity information to update solutions. Most importantly, multiple competing objective functions can be easily handled by GAs using the concept of Pareto optimality. In addition, a set of optimal tradeoff solutions can be obtained with a single algorithm run by GA.

The GAs have found increasing popularity in a wide array of civil engineering applications due to their ease of implementation and robust performance. Most early applications focused on single objective-based optimization problems only. With the fruitful advancement of multiobjective GAs in the past decade (Deb 2001), studies on GA-based multiobjective optimization have appeared recently in a variety of civil engineering problems including urban planning (Balling et al. 1999), seismic design (Liu et al. 2004, 2005b), and bridge maintenance planning (Liu et al. 1997; Miyamoto et al. 2000; Furuta et al. 2004). More detailed description of multiobjective GAs can be found in Liu et al. (2005a,b).

Application Examples

Overview

Data provided by Denton (personal communication 2002), which probabilistically describe the condition and safety deterioration processes of a large population of reinforced concrete crossheads with different maintenance strategies in the United Kingdom, are used in this study. Table 1 presents data that characterize both condition and safety profiles without maintenance. The initial performance levels and deterioration rates are assumed triangularly distributed in terms of minimum, mode, and maximum values, respectively. Table 2 provides parameters that describe different maintenance strategies and their effects on condition and safety profiles, respectively. Triangular distributions are also assumed for parameters whose uncertainties are significant.

Four different maintenance strategies indicated in Denton (personal communication 2002) are considered in this study. The time-based silane treatment reduces chloride penetration and hence decreases deterioration rates of both condition and safety. No immediate improvement or delay in deterioration, however, is produced by silane due to no correction/replacement of existing defects or deteriorated structures. There are three performance-based maintenance strategies. The "rebuild" action is launched whenever the safety index falls below a target value of 0.91. This

Table 1. Parameters for Condition and Safety Profiles without Maintenance (Data Provided by Denton, Personal Communication, 2002)

Performance index	Initial value P_0	Time of damage initiation (years) t_i	Deterioration rate (year^{-1}) α
Condition	Minimum = 0	0	Minimum = 0
	Mode = 1.75		Mode = 0.08
	Maximum = 3.50		Maximum = 0.16
Safety	Minimum = 0.91	0	Minimum = 0
	Mode = 1.50		Mode = 0.015
	Maximum = 2.50		Maximum = 0.035

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Table 2. Parameters for Condition (C) and Safety (S) Profiles with Different Maintenance Strategies (Data Provided by Denton, Personal Communication, 2002)

Parameters	Silane	Do nothing but rebuild	Minor concrete repair	Cathodic protection
Time of first application T_{pi} (years)	Design variable	When $S \leqslant 0.91$	When $C \geqslant 3.0$	When $C \geqslant 2.0$
Time interval of subsequent application T_p (years)	Design variable	When $S \leqslant 0.91$	When $C \geqslant 3.0$	Minimum=7.5 Mode=10.0 Maximum=12.5
Effects on condition index				
Improvement γ	0	Restored to $C=0$	Minimum=2.0 Mode=2.5 Maximum=3.0	0
Delay in deterioration (years) T_d	0	Minimum=10 Mode=15 Maximum=30	0	12.5
Deterioration rate during effect (year^{-1}) $\alpha-\delta$	Minimum=0 Mode=0.01 Maximum=0.03	0	Minimum=0 Mode=0.08 Maximum=0.16	0
Duration of maintenance effect (years) T_{pd}	Minimum=7.5 Mode=10.0 Maximum=12.5	Minimum=10 Mode=15 Maximum=30	0	12.5
Effects on safety index				
Improvement γ	0	Minimum=1.00 Mode=1.25 Maximum=1.50	0	0
Delay in deterioration (years) T_d	0	While $C<1.0$	While $C<1.0$	12.5
Deterioration rate during effect (year^{-1}) $\alpha-\delta$	Minimum=0 Mode=0.007 Maximum=0.018	0	Minimum=0 Mode=0.015 Maximum=0.035	0
Duration of maintenance effect (years) T_{pd}	Minimum=7.5 Mode=10.0 Maximum=12.5	While $C<1.0$	While $C<1.0$	12.5
Unit cost (k£)[a]	Minimum=0.3 Mode=39.0 Maximum=77.0	Minimum=247 Mode=7,410 Maximum=28,898	Minimum=16 Mode=3,605 Maximum=14,437	Minimum=19 Mode=2,604 Maximum=5,189

[a]Costs are indicated in pound sterling £ (GBP)≈US$1.86 as of February 2005.

action increases both condition and safety performances to those values typical of a new structure, followed by a delay in deterioration of both condition and safety; the delay in safety is terminated when the condition index is larger than 1.0. When the condition index exceeds a target value of 3.0, the minor concrete repair strategy is applied to replace all cover concrete that has visual defects but not corroded reinforcing bars. Hence, the condition is significantly improved while the safety level remains unchanged; however, the deterioration in safety is delayed until the condition index drops to 1.0. The time- and performance-based cathodic protection is carried out when the condition index reaches 2.0, and it is then applied at a regular time interval; it delays deterioration of both condition and safety for 12.5 years upon application. The four maintenance strategies considered and their corresponding probabilistic condition, safety, and cost profiles are described in Neves and Frangopol (2005).

The design variables in this study are times of first and subsequent applications, T_{pi} and T_p, of the silane treatment. They are encoded in GA as 10-bit binary strings, respectively. To start GA, an initial population of 1,000 solutions is randomly generated with a uniform distribution. Each of the subsequent generations comprises 100 offspring solutions and the nondominated elitist solutions from the previous generation. Fitness values of solutions are determined with a nondominated sorting technique that favors those closer to the global optimal front (i.e., the Pareto set) by assigning different ranks (Goldberg 1989) and then a niching strategy known as the "crowding distance" measure that discriminates solutions of the same rank (Deb 2001). A constrained binary tournament selection scheme is used to reproduce fitter solutions as well as to handle constraint-violating solutions appropriately (Deb 2001). For the encoded binary representation of each design variable, one-point crossover is applied with a probability of 80% together with one-point mutation with a probability of 10% that helps to avoid premature convergence without adverse disruption in the presence of elitism.

Although uncertainties exist in predicting condition and safety profiles, a deterministic maintenance planning optimization with silane treatment only is carried out by using modal values of the random parameters in Tables 1 and 2. This provides a direct insight into how optimal planning of maintenance interventions can be obtained in the presence of multiple conflicting objective functions, without the interference of parameter uncertainties. Effects of these uncertainties on structure performance profiles with a maintenance solution obtained from the deterministic optimization is investigated afterwards, for which "rebuild," minor concrete repair, and cathodic protection strategies are considered as well.

Optimization with Deterministic Parameters

The present GA-based procedure is now applied for optimal maintenance planning using silane. The time horizon is assumed

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50 years and a constant discount rate of 6% (Tilly 1997) is used. As practical constraints on the performance objective functions, the condition index should not exceed 3.0 and the minimum acceptable safety index value is 0.91. A total of 1,851 optimized maintenance solutions are obtained at the 40th generation. These solutions exhibit a broad tradeoff distribution over the objective functions under consideration (Fig. 3). The maintenance cost objective in general conflicts with the condition objective: the less one spends on bridge maintenance, the larger (i.e., worse) the bridge condition becomes, and vice versa. Likewise, having a safer bridge naturally requires more maintenance cost. Although condition and safety objectives are usually interact, as indicated in Table 2, they are not necessarily fully correlated because they represent different performances of a bridge component; effects of silane treatment on these two indices are not the same either.

In contrast to traditional procedures that minimize life-cycle maintenance cost only while imposing performance constraints, the significance of the present multiobjective optimization formulation for bridge maintenance planning is that it provides bridge managers with a complete set of tradeoff alternative solutions without using predefined constraints on any important performance measures (except for those ensuring practically reasonable solutions). The traditional single optimal solution may be simply viewed as a single point in Fig. 3 by enforcing specific constraints on relevant objective functions. Facing this large pool of optimized alternative solutions, decision makers now have much flexibility to select the final compromise maintenance solution, depending on the relative importance they may place on various objectives under consideration.

A total of nine representative optimized solutions, labeled *A–I*, are identified in Fig. 3 with relevant information provided in Table 3 to illustrate how a final maintenance solution could be selected through a tradeoff analysis of the 1,851 optimized solutions. In the first situation, a bridge manager sets the threshold condition level to be 3.0 at a 50 year horizon while seeking for a preferred maintenance solution that balances maintenance expenses and safety level appropriately. Solutions *A*, *B*, and *C* are associated with practically the same condition objective value of 3.0. Their time-dependent condition, safety, and cumulative maintenance cost profiles are plotted in Fig. 4, where the symbols (circle, square, and triangle) indicate application times of silane treatment of these three different maintenance solutions, respectively. It is clear that these solutions exhibit different balances of cost and safety performances: an increase in safety requires additional investment in maintenances, given the presently fixed condition objective level of 3.0.

From a different viewpoint, another set of three solutions, denoted as *D*, *E*, and *F*, are investigated. These solutions all have a safety index value of 1.21 at 50 years (see Table 3). Their time-dependent condition, safety, and cost profiles are given in Fig. 5. In this situation, a bridge manager faces another tradeoff between maintenance cost and condition level: an improvement in one objective function has to be accompanied by sacrificing the other. Specifically, if one selects solution *F* instead of *E*, a saving of 21.83 k£ or 27.1% in maintenance expenses is achieved; the condition objective value, however, is increased from 2.44 to 2.90, representing an 18.9% decrease in condition performance. It is reiterated that, if the maintenance cost is used as the only objective function as in traditional optimal maintenance planning procedures, solution *F* would be the single optimal solution provided the constraints on condition and safety are prescribed as 2.90 and 1.21, respectively; however, the bridge manager would have no basis to judge whether it is worth spending more on maintenances

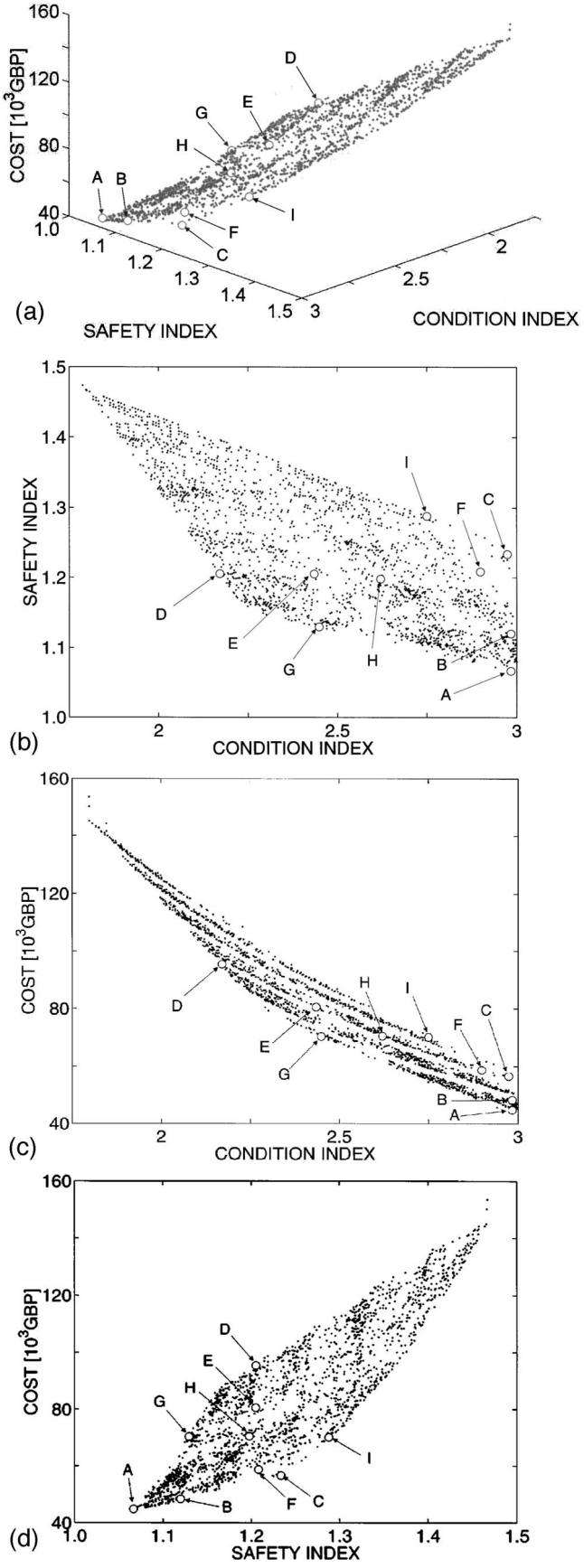


Fig. 3. Tradeoff among different objective functions for 1,851 optimized maintenance solutions using silane: (a) three-dimensional plot; (b)–(d): two-dimensional projections

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Table 3. Nine Representative Optimized Silane Maintenance Solutions

Solution ID	Time of first application (years) T_{pi}	Time interval of subsequent applications (years) T_p	Present value of 50 year cumulative cost (k£)	Highest (worst) condition index value during 50 years	Lowest (worst) safety index value during 50 years	Note
A	9.65	10.30	44.79	2.98	1.07	Condition index values at year 50 are almost equal.
B	11.85	7.70	48.34	2.98	1.12	
C	14.45	5.20	56.57	2.97	1.23	
D	0	8.40	95.42	2.17	1.21	Safety index values at year 50 are equal.
E	4.55	7.50	80.54	2.44	1.21	
F	12.55	5.90	58.71	2.90	1.21	
G	3.00	9.95	70.33	2.45	1.13	Present values of 50 year cumulative cost are almost equal
H	7.60	7.00	70.47	2.62	1.20	
I	12.00	4.90	70.09	2.75	1.29	

than that of the traditional single optimal solution in order to achieve a better condition level during the designated service life.

The third situation is unique in that, other than minimizing the life-cycle maintenance cost as most existing maintenance planning optimization tools do, a bridge manager now fixes the maintenance cost to, for instance, about 70 k£ while examining the optimized tradeoff between condition and safety objectives. Three such optimized scenarios are also listed in Table 3 and are referred to as solutions *G*, *H*, and *I*, respectively. Fig. 6 provides time-dependent condition, safety, and cost profiles of the three maintenance solutions. This represents a practical situation where the available budgets are less than the expenses needed for the optimal maintenance solution from the traditional life-cycle cost minimization. As a result, a bridge manager is interested in seeking for an alternative maintenance solution that can strive for the most desired balance between bridge condition and safety objectives under financial constraints. This has not been systematically studied in the literature of civil infrastructure maintenance planning.

Effects of Parameter Uncertainties

In the above application examples of multiobjective maintenance planning optimization, modal values of random parameters in the deterioration models have been used for computing condition, safety, and cost profiles. It is expected that, in the presence of

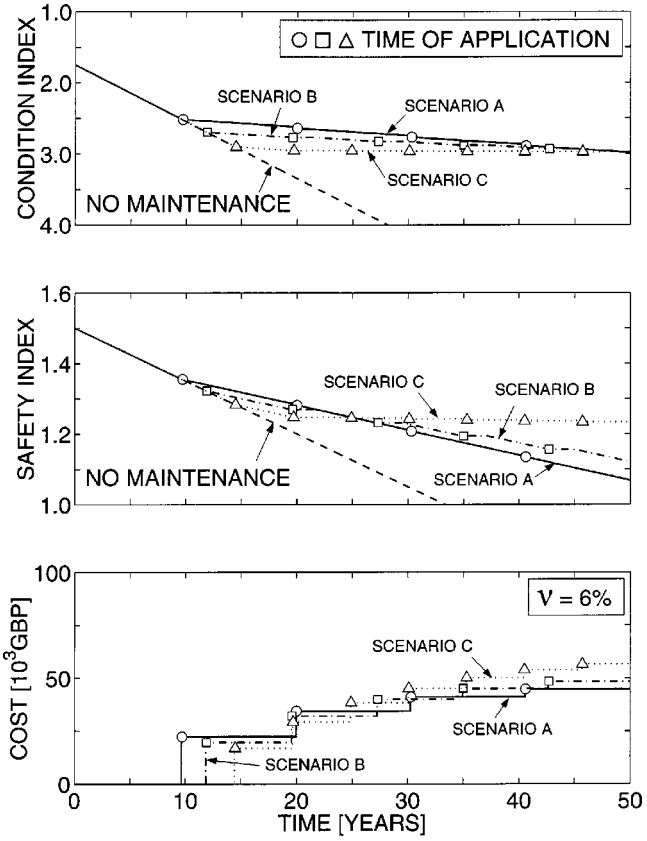


Fig. 4. Profiles with different optimized application times of silane associated with condition indices close to 3.00 at year 50

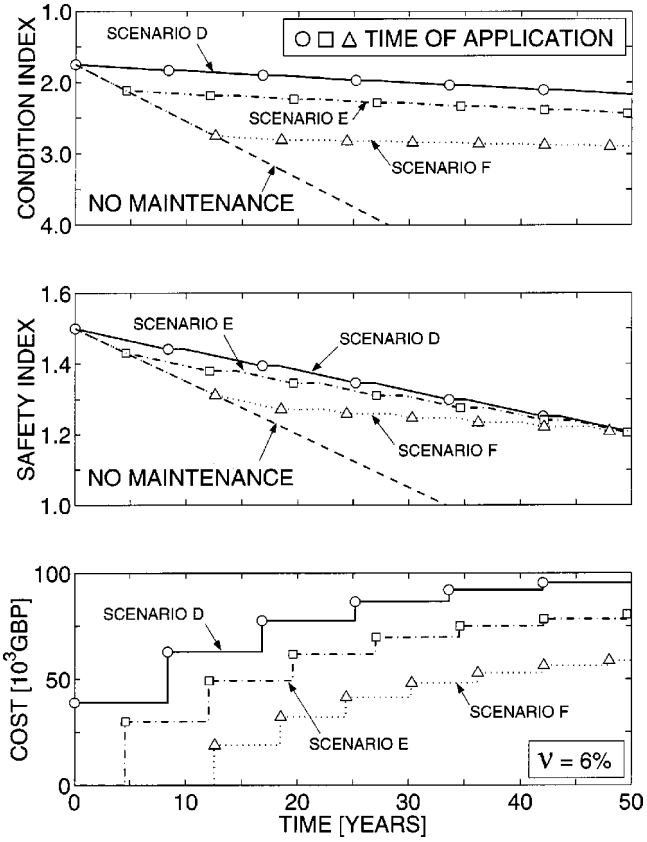


Fig. 5. Profiles with different optimized application times of silane associated with same safety indices of 1.21 at year 50

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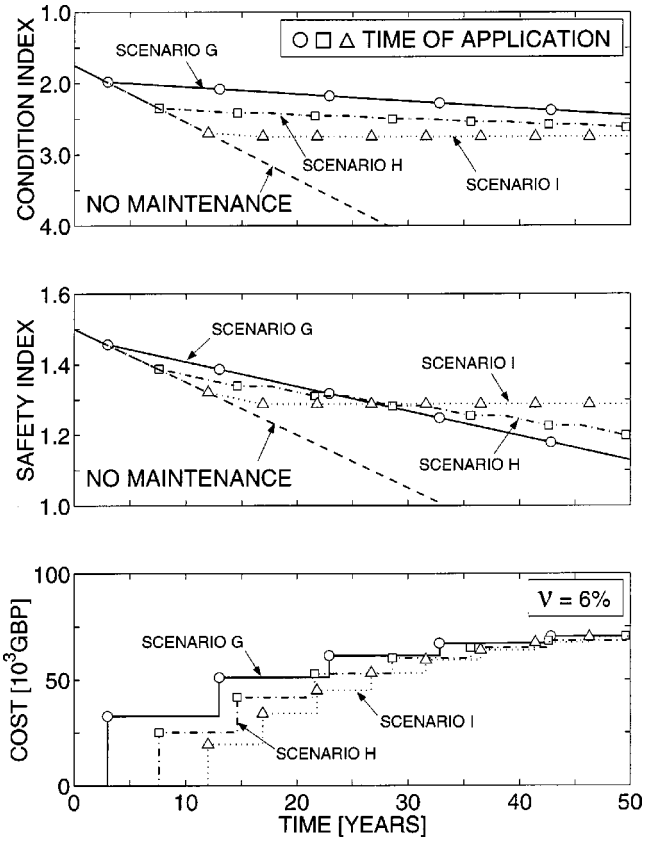


Fig. 6. Profiles with different optimized application times of silane associated with cumulative 50 year maintenance costs close to 70 k£

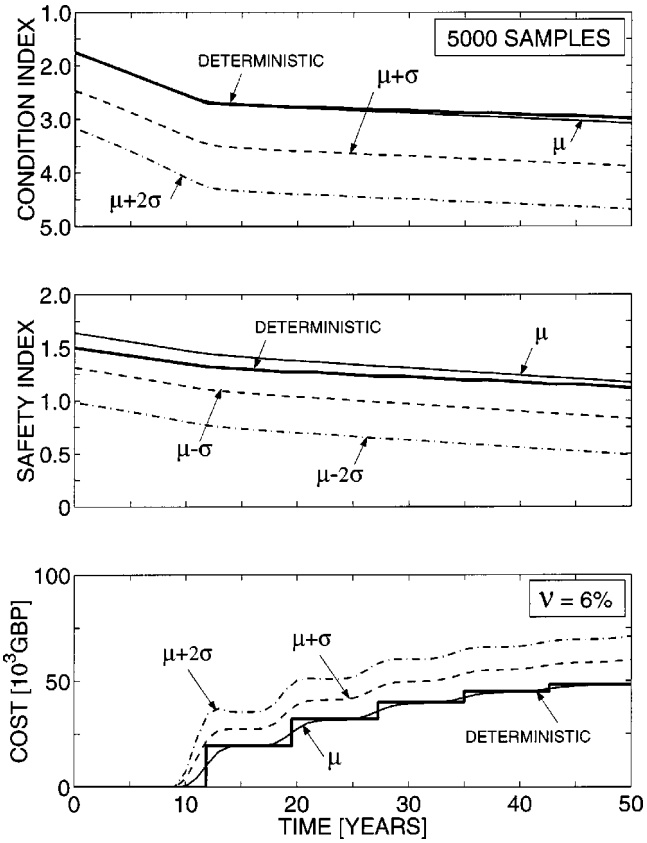


Fig. 7. Profiles with solution *B* using deterministic analysis and Monte Carlo simulation

parameter uncertainties, the calculated objective values will vary accordingly. In this section, Monte Carlo simulation is carried out with a sample size of 5,000 to account for these uncertainty effects. The relevant random parameters all follow triangular distributions with minimum, mode, and maximum values provided in Tables 1 and 2, respectively.

For illustration purposes, times of first and subsequent cyclic application of silane, T_{pi} and T_p, are assumed triangularly distributed with modal values of 11.85 and 7.70 years, respectively, which have been used deterministically for solution *B* as provided in Table 3, and a coefficient of variation equal to 0.0816. Fig. 7 plots the condition, safety, and cost profiles in terms of their sample mean μ, sample mean ± 1 sample SD $\mu+\sigma$ (or $\mu-\sigma$), and sample mean ± 2 sample SDs $\mu+2\sigma$ (or $\mu-2\sigma$), where SD =standard deviation. The profiles obtained deterministically using modal parameter values are also plotted as a comparison. It is seen that deterministic profiles in general match sample mean profiles, respectively. The dispersions of each profile, however, are very significant. Fig. 8 shows histograms of the condition index, safety index, and maintenance cost evaluated at year 50, respectively.

The implication is that deterministic optimization based on modal parameter values may lead to optimized maintenance solutions with objective values exceeding predefined limits if percentile other than mean objective values are used in the optimization process. For example, if the mean +1 SD values (i.e., 84th percentile under normal distribution assumption) are adopted instead, solution *B* leads to a condition index value of 3.88 and a safety index value of 0.83 at year 50, which exceed the acceptable condition index value of 3.0 and acceptable safety index value of 0.91.

Application of Performance-Based Maintenance Strategies

It should be noted that, in the Monte Carlo simulation, the condition and/or safety level at some time instant may be too low and other maintenance strategies ("rebuild," minor concrete repair, cathodic protection) as listed in Table 2 are therefore necessary when respective target performance values are reached. These performance-based maintenance strategies are first applied individually to illustrate their respective effects on structure performances and then applied in selected combinations to show the possibility of obtaining more efficient maintenance solutions.

Fig. 9 presents condition, safety, and cost profiles, in terms of sample mean values obtained by Monte Carlo simulation, with each individual maintenance strategy. They are obtained by using the methodology described in Frangopol and Neves (2003). The associated objective values with each strategy are provided in Table 4. It is observed that minor concrete repair provides the best condition performance over the entire service life; on the other hand, it also leads to the lowest safety level among all different maintenance strategies. Cathodic protection yields excellent condition and safety performances but with the highest maintenance cost over the entire service life. The "rebuild" strategy enhances both structure condition and safety; the lowest performances occur roughly in the middle of service life. Because silane can neither improve nor delay deterioration of condition and safety, it

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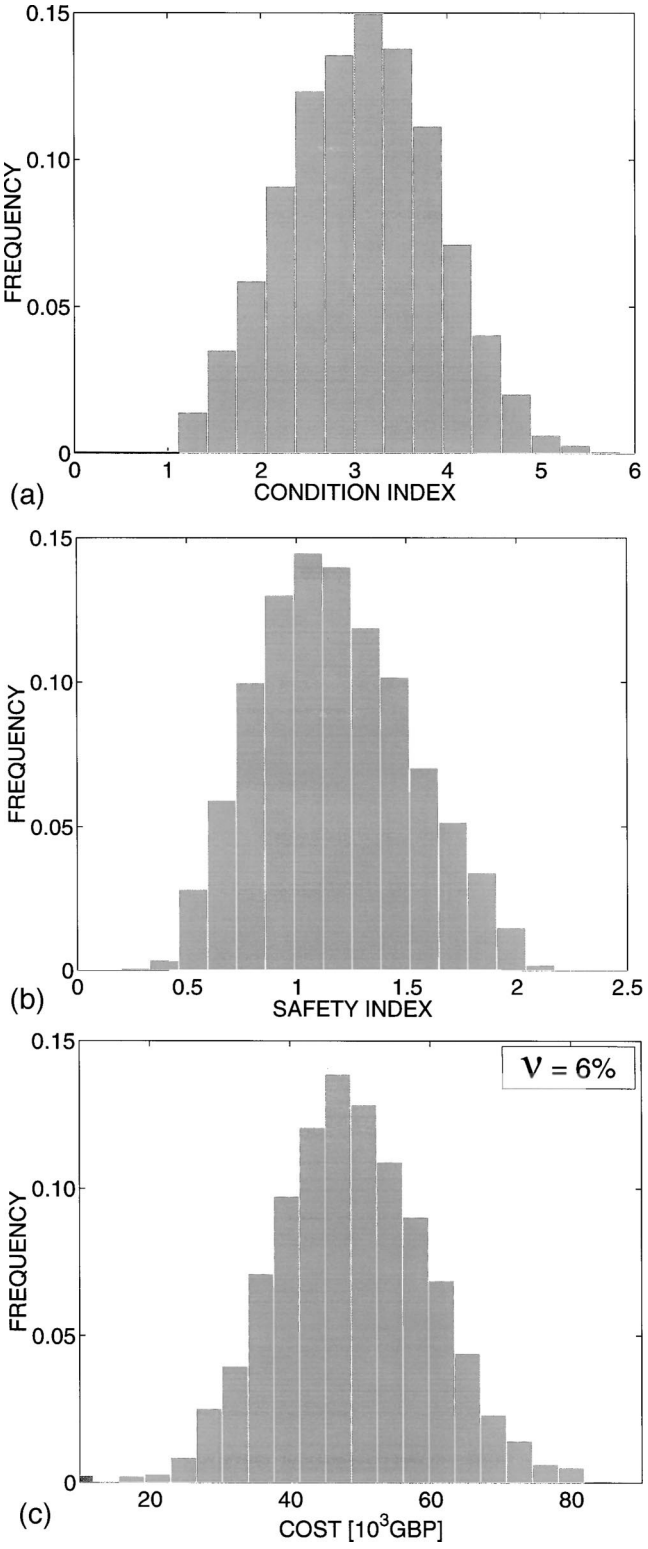


Fig. 8. Histograms of performance indices of solution *B* at year 50 using Monte Carlo simulation: (a) condition index; (b) safety index; and (c) present value of cumulative maintenance cost

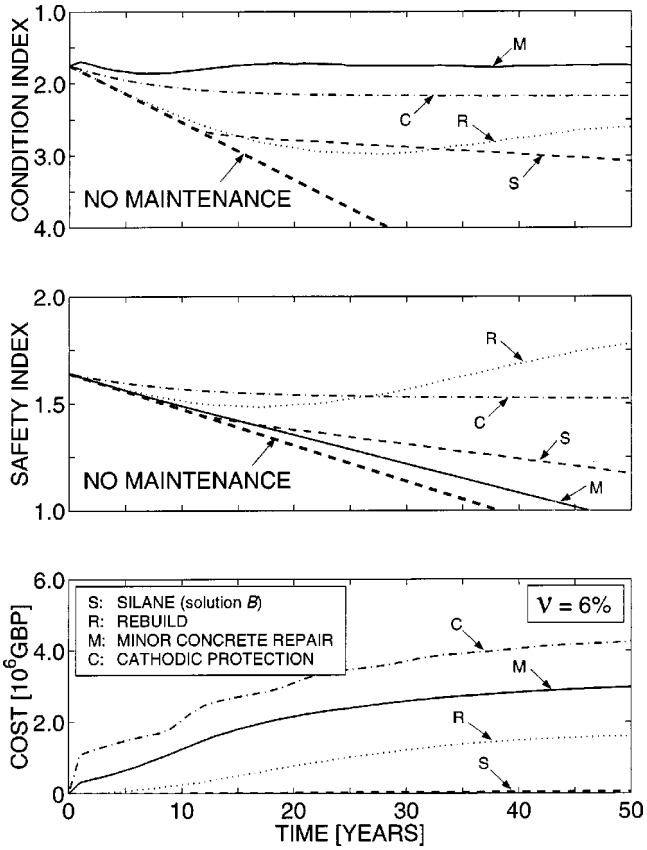


Fig. 9. Sample mean profiles with different individual maintenance strategies using Monte Carlo simulation

is necessary that silane be combined with one or more performance-based maintenance strategies so that more economical maintenance solutions can be obtained. The methodology of combining probabilistic maintenance actions is described in Petcherdchoo et al. (2004).

Next, silane solution *B* is applied with "rebuild," minor concrete repair, and cathodic protection, respectively. The resulting sample mean profiles are reported in Fig. 10 and the associated objective values with each maintenance combination are provided in Table 5. By comparing Figs. 9 and 10 as well as Tables 4 and 5, one can find that the combination of silane with another individual performance-based maintenance strategy generally improves the structure performance when no silane treatment is added. For example, compared to use of "rebuild" alone, by combining silane and "rebuild" strategies, the condition objective value is reduced from 2.98 to 2.59 while the safety objective value is increased from 1.17 to 1.47; at the same time, the cost objective also favorably drops from 1,606.5 to 817.2 k£. Although there is no change in condition performance, the combination of silane and minor concrete repair improves, compared to use of minor concrete repair alone, the safety objective value from 0.95 to 1.21 and also reduces the sample mean cost from

Table 4. Sample Mean Objective Values with Different Individual Maintenance Strategies Using Monte Carlo Simulation

Maintenance	Highest (worst) condition index value during 50 years	Lowest (worst) safety index value during 50 years	Present value of 50 year cumulative cost (k£)
Silane	3.08	1.17	48.7
Rebuild	2.98	1.48	1,606.5
Minor concrete repair	1.86	0.95	2,978.8
Cathodic protection	2.18	1.52	4,250.6

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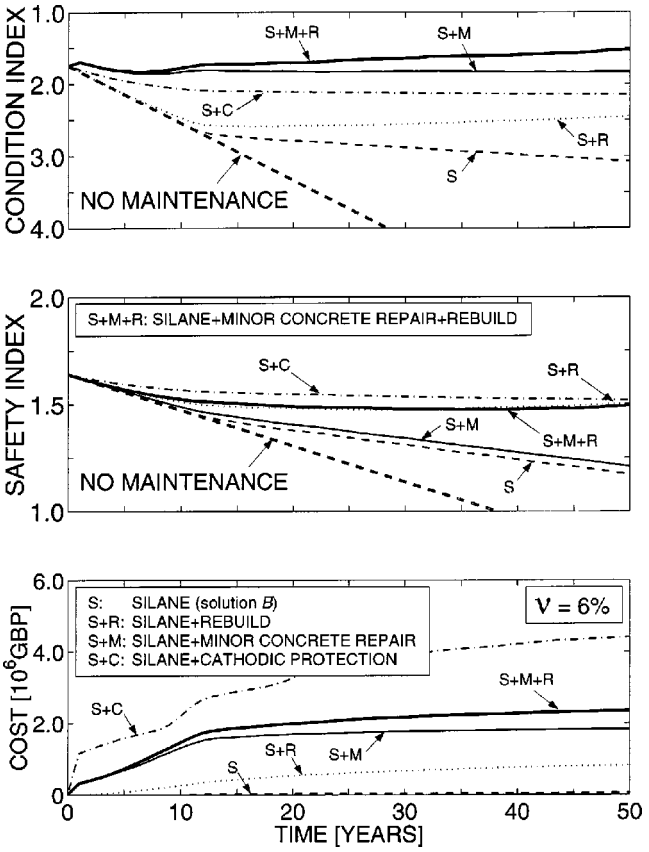


Fig. 10. Sample mean profiles with different maintenance strategy combinations using Monte Carlo simulation

2,978.8 to 1,831.5 k£, indicating a saving of 38.5%. However, the combination of silane and cathodic protection practically does not improve condition or safety level compared to use of cathodic protection alone.

Because the "rebuild" and minor concrete repair strategies excel in improving safety and condition performances, respectively, it is interesting to combine silane, minor concrete repair, and rebuild together so that one has the opportunity to improve both condition and safety levels simultaneously. As clearly shown in Fig. 10 and Table 5, the resulting condition and safety objective values are 1.83 and 1.47, respectively. In this case, as a tradeoff, a bridge manager has to invest more for maintenances in this case. Whether this extra cost is worthwhile will be evaluated through a decision making process.

Table 5. Sample Mean Objective Values with Different Maintenance Strategy Combinations Using Monte Carlo Simulation

Maintenance combination	Highest (worst) condition index value during 50 years	Lowest (worst) safety index value during 50 years	Present value of 50 years cumulative cost (k£)
Silane + rebuild	2.59	1.47	817.2
Silane + minor concrete repair	1.86	1.21	1,831.5
Silane + cathodic protection	2.15	1.52	4,405.4
Silane + minor concrete repair + rebuild	1.83	1.47	2,350.0

Conclusions

In this paper, genetic algorithm based multiobjective optimization techniques have been applied for maintenance planning of deteriorating bridges. Structure condition, safety, and cumulative life-cycle maintenance cost have been considered as separate objective functions subject to simultaneous optimization. Unlike traditional bridge management system tools aiming to seek for a single maintenance solution that minimizes life-cycle maintenance cost alone, the present automated procedure produces a large pool of alternative maintenance solutions that establishes optimized tradeoff among all competing merit objectives. As a result, bridge managers have a much broader perspective of the entire scope of optimal maintenance solution options. Through an explicit tradeoff analysis, bridge managers are able to actively choose the final compromise maintenance solution that has desirable condition and safety levels with balanced life-cycle maintenance cost, other than passively accepting performance levels implied by the single solution resulting from the traditional long-term cost minimization only.

Monte Carlo simulation results have shown that uncertainties associated with the deterioration and maintenance processes may have significant effects on estimating performance profiles as well as life-cycle maintenance cost. Besides the time-based silane treatment used in the present optimization process, other performance-based maintenance strategies, including "rebuild," minor concrete repair, and cathodic protection may also be considered in order to obtain more practical and cost-effective maintenance solutions.

The writers have considered probabilistic structure performance profiles in bridge maintenance planning optimization (Liu and Frangopol 2004) and carried out annual maintenance prioritization using a variety of maintenance strategies (Liu and Frangopol 2005a). In addition to the above project-level bridge maintenance management for a group of similar structures, the writers have also investigated the network-level bridge maintenance management (Liu and Frangopol 2005b), in which limited resources are prioritized to bridges that form the critical nodes of a highway network in order to cost effectively maintain the satisfactory long-term performance of the entire network.

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